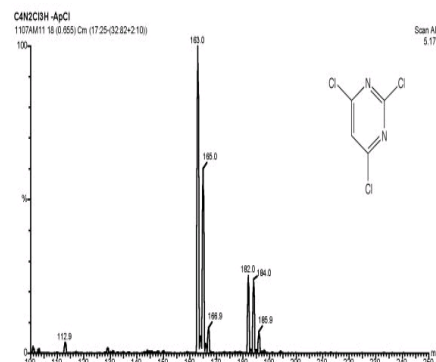


Determining Chemical Formulas

Law of Constant Composition: compounds contain elements in fixed proportions....
Always the SAME!

Equipment for determining the formula (proportions):

1. **Mass Spectrometer (CSI):** can determine the molar mass of many compounds by vapourizing a small sample, ionizing it (giving it a charge) and deflecting the charged gas in a magnetic field. The amount of deflection gives the mass.
2. **Combustion Analyzer:** Determines amount of C, H, O and N by burning the substance in pure oxygen and analyzing the products.



Empirical Formula

- Gives the lowest ratio of atoms in a compound, but may not tell the exact number of each type of atom

VS.

Molecular Formula

- Indicates the actual numbers of atoms in one molecule of a compound

	Molecular	Empirical
Hydrogen peroxide		
Sugar		

Percentage composition: percentage, by MASS, of each element in a compound

Examples (p111):